

ANTHROPIC ACADEMY — COURSE SUMMARIES

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PRODUCT

Product

DEVELOPER

Developer

AI FLUENCY

AI Fluency

01 Claude 101

Master everyday Claude usage across chat, projects, artifacts, and enterprise tools.

PRODUCT

- What Claude is & having your first conversation; prompting tips for better results
- Projects: organise long-running work with shared context across conversations
- Artifacts: generate and iterate on documents, code, and structured outputs
- Skills, tool connections, enterprise search, and Research mode for deep dives
- Role-specific use-cases; Claude desktop app (Chat / Cowork / Code tabs)

02 Claude Platform 101

From first API call to production agent loops — built for developers.

DEVELOPER

- API requests from first principles: structure, tokens, context window limits
- Model selection: Opus / Sonnet / Haiku cost-latency trade-offs on real examples
- Build the agent loop by hand (act → observe → decide), then simplify with SDK Tool Runner
- Extended thinking, prompt caching, workspaces & spend limits via Console Workbench
- Built-in tools (web search, code execution), Skills, MCP servers, managed agents

03 Introduction to Claude Cowork

Run multi-step tasks over your real files and apps with Cowork.

PRODUCT

- Cowork task loop: how Claude plans, acts on files/apps, and reports back
- Setting up and managing plugins to extend Cowork's reach to third-party services
- File & research workflows: upload docs, trigger web research, combine outputs
- Scheduling recurring tasks and managing async autonomous workflows
- Steering techniques: when to intervene, approve, or redirect mid-task

04 Claude Code in Action

Integrate Claude Code into your daily development workflow.

DEVELOPER

- AI coding-assistant architecture: how Claude reads and edits code via tool integration
- Tool-use system: combining multiple tools for complex multi-step programming tasks
- Context management: referencing project resources and keeping sessions relevant
- Extending via MCP servers and GitHub integration for real-world workflows
- Explore → Plan → Code → Commit cycle; CLAUDE.md for project-level config

05 AI Fluency: Framework & Foundations

4D framework for effective, efficient, ethical, and safe AI collaboration.

AI FLUENCY

- Delegation: identify which tasks belong to AI vs. human judgment
- Description: craft clear, specific prompts that yield high-quality outputs
- Discernment: critically evaluate AI responses — spot errors, bias, and gaps
- Diligence: verify outputs, maintain accountability, and iterate responsibly
- Where AI capability boundaries lie; ethical use and bias-awareness habits

06 Building with the Claude API

Full API lifecycle from authentication to production AI-powered systems.

DEVELOPER

- API key setup, request config, message structure, and response parsing
- Advanced prompting: system prompts, multi-turn conversations, streaming
- Tool integration (function calling) and RAG patterns for knowledge retrieval
- Multimodal inputs: processing text, images, and documents in one request
- Automated workflows, architectural patterns, and hands-on exercises throughout

07

Intro to Model Context Protocol

Build MCP servers & clients in Python; connect Claude to any external service.

DEVELOPER

- MCP architecture: shifts tool-definition burden from app code to dedicated servers
- 3 core primitives — Tools (model-controlled), Resources (app-controlled), Prompts (user-controlled)
- Build servers with Python SDK decorators; test live with built-in MCP Server Inspector
- Define templated resources, implement MIME-typed resource reading in clients
- Autocomplete & context injection patterns; transport-agnostic communication model

09

AI Capabilities & Limitations

Build a working mental model of how generative AI behaves — and why it fails.

AI FLUENCY

- How LLMs generate text: pattern completion, not retrieval or reasoning chains
- Capability spectrum: language, summarisation, code — vs. precise math, facts, memory
- Types of unexpected outputs: hallucination, misalignment, over-confidence
- Locate a task on the capability-to-limitation continuum; choose targeted fixes
- Complements the 4D Framework — the machine properties behind each human competency

08

MCP: Advanced Topics

Production MCP patterns: sampling, notifications, transports, and scaling.

DEVELOPER

- Sampling: MCP servers request LLM calls through clients — shifts AI cost to client
- Progress & logging notifications with context objects for long-running operations
- Roots-based file access: grant servers directory scope with security boundaries
- Stdio vs StreamableHTTP (SSE) transports: initialization handshake & dual-connection
- Stateless HTTP for horizontal scaling with load balancers vs. stateful trade-offs